Operation Ember Ascent: Mustafar Two-Stage Sounding Rocket

STAC WARS PS1 | Team: Nova | Software: OpenRocket 24.12

1. Design approach

Body material: **cardboard** (mass limit 10 kg). The rocket is a two-stage design (sustainer + booster), 144 cm long, 5 cm maximum diameter, with a 1.00 kg scientific payload modelled as a normal mass component (no mass overrides). Total launch mass with motors is 2926 g.

I first set the Mustafar environment, then chose motors. My first motor choice (J435WS + J350W, 1441 N·s) gave about 6100-6900 m, far above the 4 km target. I reduced impulse to an AeroTech I285R (sustainer, plugged delay) and an AeroTech I366R-4 (booster), about 959 N·s in total (limit 2800 N·s). I then fine-tuned apogee with an added ballast mass component (333 g). The booster motor delay was shortened from 14 s to 4 s so the stages separate shortly after burnout; with the long delay the stack coasted, tilted into the wind and the sustainer reached apogee with a large sideways speed.

Item	Final design
Motors	Booster AeroTech I366R-4; sustainer AeroTech I285R (plugged)
Total impulse	about 959 N·s (limit 2800)
Length / mass	144 cm (limit 220) / 2926 g with motors (limit 10 kg)
Payload	1.00 kg
Stability (full stack)	3.0 cal in the OpenRocket design view (CG 71.2 cm, CP 86.2 cm); 4.31 cal in flight at t = 2 s; sustainer alone 5.14 cal
Apogee	4020 m (target 3900-4100 m)
Max velocity	301 m/s (Mach 0.54)

2. Mustafar launch conditions

Atmosphere set manually (ISA off): 426.85 C (700 K), 750 mbar (75 kPa). OpenRocket derives density from these, giving about 0.37 kg/m^3, which matches the briefed 0.374. Wind: 10 m/s average, direction 270 deg, turbulence 2%, standard deviation 0.5 m/s. Launch altitude 0 m, 250 cm launch rod, vertical launch.

Limitation: OpenRocket 24.12 has no gravity setting that we could find, so all runs use its built-in gravity (9.78-9.79 m/s^2, confirmed in the exported flight data), not Mustafar's 6.5 m/s^2. The effect of lower gravity below is therefore a prediction that was not simulated.

3. Recovery system

Sustainer parachute: deploys at **apogee**. Booster parachute: deploys at the booster's own apogee, which avoids opening at high speed. Both sit in their own tube sections. The 1.00 kg payload (1000 g) is carried inside the sustainer and is recovered with it under the sustainer parachute, which deploys at apogee. Sustainer descent rate is 6.86 m/s and booster descent rate is 5.62 m/s (average vertical velocity over the last 60 s of descent), both inside the 4-8 m/s requirement. Note: the sim still shows a 'deployment at high speed' warning for the sustainer chute (30.8 m/s) because the rocket is travelling sideways in the 10 m/s wind at apogee; with no wind the same event is at 0.64 m/s.

4. Earth vs Mustafar comparison

The same final rocket was flown with one environmental change at a time, starting from Earth standard conditions (101.3 kPa, 15 C, 10 m/s wind).

Run	Setting	Air density (kg/m3)	Apogee (m)	Max vel (m/s)	Time to apogee (s)	Stability at t=2 s (cal)	Descent, sustainer / booster (m/s)	Sustainer drift (m)
Earth_Baseline	ISA, 10 m/s wind	1.225	2613	265	23.5	4.33	3.81 / 3.12	5678
Temp_Only	700 K, 101.3 kPa	about 0.50	3625	294	28.1	4.32	5.90 / 4.84	5283
Pressure_Only	15 C, 75 kPa	about 0.91	2975	277	25.3	4.30	4.43 / 3.63	5336
Wind_Only (no wind)	700 K, 75 kPa, 0 m/s	about 0.37	4136	302	30.3	4.40	6.87 / 5.63	12
Mustafar_Final	700 K, 75 kPa, 10 m/s	about 0.37	4020	301	29.9	4.31	6.86 / 5.62	4943

Densities are calculated from rho = P/(R T). Stability is the in-flight value at t = 2 s. Descent is the mean vertical speed over the last 60 s. Gravity could not be changed (see section 2).

5. Predictions and results

Change	Prediction	Result	Held?
Thinner air (density 1.225 to 0.37)	Less drag, higher apogee	Earth 2613 m to Mustafar 4020 m (+1407 m). Temperature alone +1012 m, pressure alone +362 m; the two effects roughly add (1374 m).	Yes
Higher temperature	Speed of sound rises (about 340 to about 530 m/s), so Mach number falls	Mustafar max velocity 301 m/s is Mach 0.54; Earth 265 m/s is about Mach 0.78 (estimated from a 340 m/s speed of sound).	Yes
Lower pressure / density	Faster parachute descent for the same canopy	Sustainer descent 3.81 m/s on Earth, 6.86 m/s on Mustafar (about 1/sqrt(density), as expected). Temperature-only 5.90, pressure-only 4.43 m/s.	Yes
10 m/s side wind	Weathercocking, lower apogee, downrange drift	Removing wind raises apogee from 4020 to 4136 m (+116 m); sustainer chute deployment speed falls from 30.8 to 0.64 m/s. Sustainer drift is 4943 m with wind and 12 m without.	Yes
Lower gravity (6.5 m/s2)	Lower weight, longer climb and higher apogee	Not simulated: OpenRocket 24.12 has no gravity input	Untested

6. Conclusion

The final design meets the mission requirements: apogee 4020 m, length 144 cm, mass 2926 g with cardboard body, impulse about 959 N·s, payload 1.00 kg, descent rates in the 4-8 m/s range, and recovery by parachute for both stages. Thinner, hotter air raised apogee by about 1400 m relative to Earth, and the 10 m/s wind reduced apogee and caused the high-speed deployment warning. Gravity could not be changed, so that prediction remains untested. Stability was 3.0 cal for the full stack in the design view (4.31 cal in flight at t = 2 s) and 5.14 cal for the sustainer alone; no minimum was specified in the brief.